



An integrated multi-diagnostic visualization platform for KSTAR tokamak

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ABSTRACT

An integrated visualization platform named PRISM (Plasma Research Integrated System for Multi-diagnostics) has been developed for the KSTAR tokamak. Comprehensive plasma characterization requires examining data from multiple diagnostic systems, but these data are typically accessed through separate tools with different interfaces. Comparing measurements across diagnostics also requires coordinate transformation to a common magnetic geometry, which adds additional steps to the analysis workflow. PRISM addresses these issues by integrating multiple diagnostics under a unified interface with built-in equilibrium mapping and data quality controls. The platform enables simultaneous visualization of profiles and time traces from multiple shots, time points, and spatial channels, facilitating systematic parameter scans and cross-diagnostic validation. PRISM also provides two-dimensional radiation profile visualization with equilibrium overlay and visible camera image sequence analysis. For magnetohydrodynamic fluctuation studies, spectrogram generation, and toroidal mode number spectrum analysis are included. The modular architecture allows straightforward extension to additional diagnostics. PRISM has been deployed on the KSTAR experimental server and is actively used by researchers for experimental data assessment.

1. Introduction

Tokamak experiments employ a wide range of diagnostic systems to measure plasma parameters such as temperature, density, radiation, and magnetic fluctuations. Researchers routinely need to compare these measurements, but doing so is not straightforward as each diagnostic has its own data format and access interface. Cross-diagnostic comparison further involves coordinate transformation to a common magnetic geometry through equilibrium reconstruction, since different diagnostics measure at different spatial locations. While an increasing number of diagnostics improves plasma characterization, these challenges become more pronounced.

The KSTAR tokamak is equipped with more than 50 diagnostic systems for comprehensive plasma measurements [1], including Charge Exchange Spectroscopy (CES) for ion temperature and rotation [2], as well as density [3], X-ray Imaging Crystal Spectrometer (XICS) for ion temperature and rotation [4], Thomson Scattering (TS) for electron temperature and density [5], Electron Cyclotron Emission (ECE) radiometry for electron temperature profiles [6], Electron Cyclotron Emission Imaging (ECEI) for two-dimensional electron temperature fluctuations [7–11], Two-Color Interferometer (TCI) for line-integrated electron density [12], Motional Stark Effect (MSE) diagnostics for magnetic pitch angle, safety factor, and current density profiles [13,14], Beam Emission Spectroscopy (BES) using neutral beam (NB) injection for density fluctuations [15], Mirnov coil arrays for magnetic fluctuations [16], Infrared Video Bolometer (IRVB) for plasma radiation

profiles [17], and In-Vessel Inspection System (IVIS) TV cameras for plasma imaging [18]. Managing data from this diverse set of diagnostics motivates the development of an integrated visualization platform.

At KSTAR, diagnostic data visualization has relied on several separate tools. jScope [19], a general-purpose MDSplus waveform viewer, has been the primary tool for displaying time-series signals but does not offer diagnostic-specific processing such as equilibrium mapping or profile coordinate transformation. A standalone tool for Mirnov coil and ECE analysis provided spectrogram visualization as well as toroidal mode number identification for magnetohydrodynamic (MHD) fluctuation studies. Beyond these, researchers typically wrote individual scripts for their specific needs, resulting in a fragmented workflow.

In the 2024 KSTAR campaign, the present author developed CESviewer, an internal visualization tool distributed to KSTAR experimental participants for ion temperature as well as toroidal rotation profiles measured by CES with Equilibrium Fitting (EFIT) coordinate mapping [20]. Building on this foundation, PRISM (Plasma Research Integrated System for Multi-diagnostics) was developed to extend the approach to multiple KSTAR diagnostics under a unified interface with built-in EFIT mapping and automatic data quality controls. The platform enables simultaneous visualization of profiles and time traces from multiple shots, time points, and spatial channels, facilitating systematic parameter scans as well as cross-diagnostic validation. For

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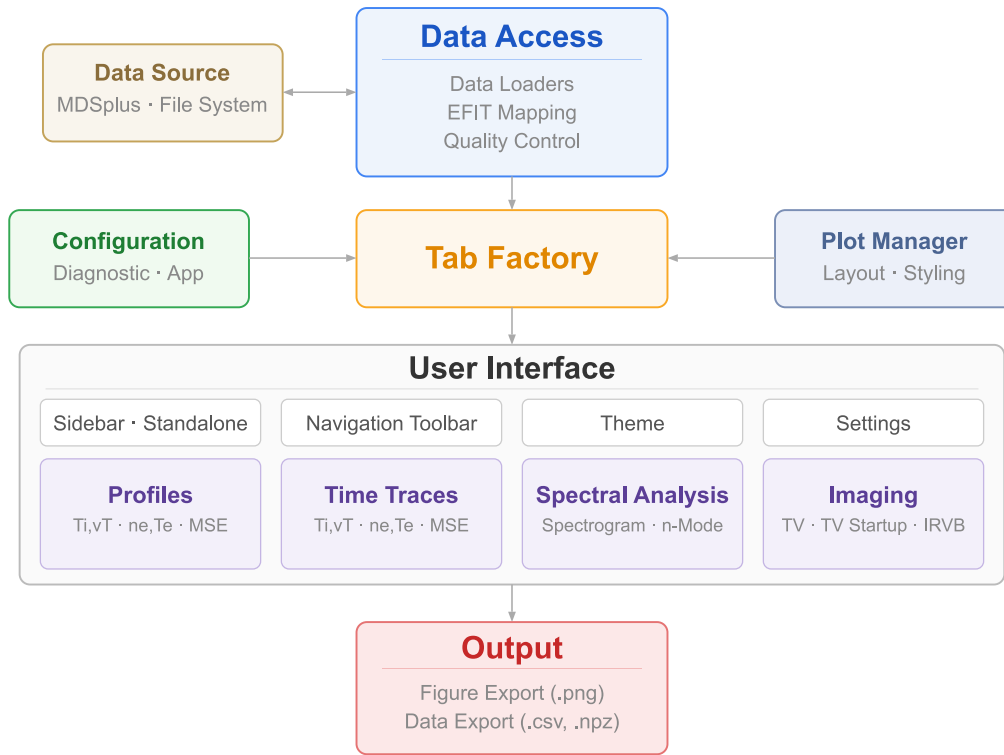


Fig. 1. System architecture of PRISM. Data from the MDSplus server and file system is retrieved through the Data Access layer with built-in EFIT mapping and quality control. The Tab Factory assembles viewer tabs using the Configuration and Plot Manager components. The User Interface organizes eleven viewer tabs into four categories (Profiles, Time Traces, Spectral Analysis, and Imaging) via sidebar navigation, with a standalone selector mode. Processed data can be exported as figures (.png) or data files (.csv, .npz).

two-dimensional measurements, PRISM provides radiation profile visualization from IRVB with equilibrium overlay, as well as visible camera image analysis. The MHD fluctuation analysis is extended to support ECE, ECEI, BES, and TCI in addition to Mirnov coils, providing spectrogram generation and toroidal mode number (n) spectrum analysis. The modular architecture allows straightforward extension to additional diagnostics. While PRISM includes built-in analysis capabilities such as spectrogram generation as well as toroidal mode number identification, the platform is primarily designed for rapid data visualization; for detailed quantitative analysis and profile fitting, dedicated tools are used.

This paper is organized as follows. Section 2 describes the system architecture and design patterns. Section 3 presents the supported diagnostics with visualization examples. Section 4 discusses key features including data quality control, multi-shot comparison, and equilibrium mapping. Section 5 compares PRISM with existing tools and discusses applications. Section 6 summarizes the work and outlines future development plans.

2. System architecture

Fig. 1 shows the overall system architecture of PRISM, which is designed with a modular architecture that separates concerns into seven distinct components: Data Sources, Data Access (diagnostic-specific loaders with EFIT mapping and quality control), Configuration, Tab Factory, Plot Manager, User Interface, and Output. This separation enables independent development and testing of each component, as well as straightforward extension to new diagnostics. PRISM is implemented in Python using PySide6 (Qt 6) for the graphical user interface and matplotlib, a Python plotting library, for data visualization. The software is managed using Git version control with semantic versioning. PRISM is deployed on the KSTAR experimental server with symbolic links enabling system-wide access through the `prism` command.

The graphical user interface of PRISM is shown in Fig. 2. A categorized sidebar on the left groups diagnostic viewers into Profiles, Time Traces, Spectral Analysis, and Imaging categories. The center panel displays the plot canvas, while the right panel contains control sections for data loading, time selection, customizable plotting options and EFIT mapping. The bottom toolbar provides standard navigation functions for zooming, panning, axis range adjustment, and saving figures, as well as a light/dark theme toggle and a link to the KSTAR diagnostics manual. The specific control layout varies by tab type, with profile viewers, time trace viewers, and analysis viewers each offering relevant options. Profile tabs additionally provide per-channel visibility control and MDSplus node identification that overlays node names directly on the plot.

2.1. Design patterns

The software employs the factory design pattern — a creational pattern in which a dedicated class is responsible for instantiating objects without specifying their concrete types — for tab creation. The `TabFactory` class maintains mappings between diagnostic names and their corresponding loader classes, parser classes, and tab classes. When a new diagnostic tab is requested, the factory instantiates the appropriate components and assembles them into a functional tab. This approach eliminates the need for conditional logic scattered throughout the codebase and centralizes the tab creation logic in a single location.

All diagnostic tabs inherit from an abstract `BaseTab` class that defines the common interface and shared functionality. This includes methods for creating selection listboxes, EFIT mapping controls, axis control panels, and data export functions. Each specific diagnostic tab implements only the diagnostic-specific logic while reusing the common infrastructure.

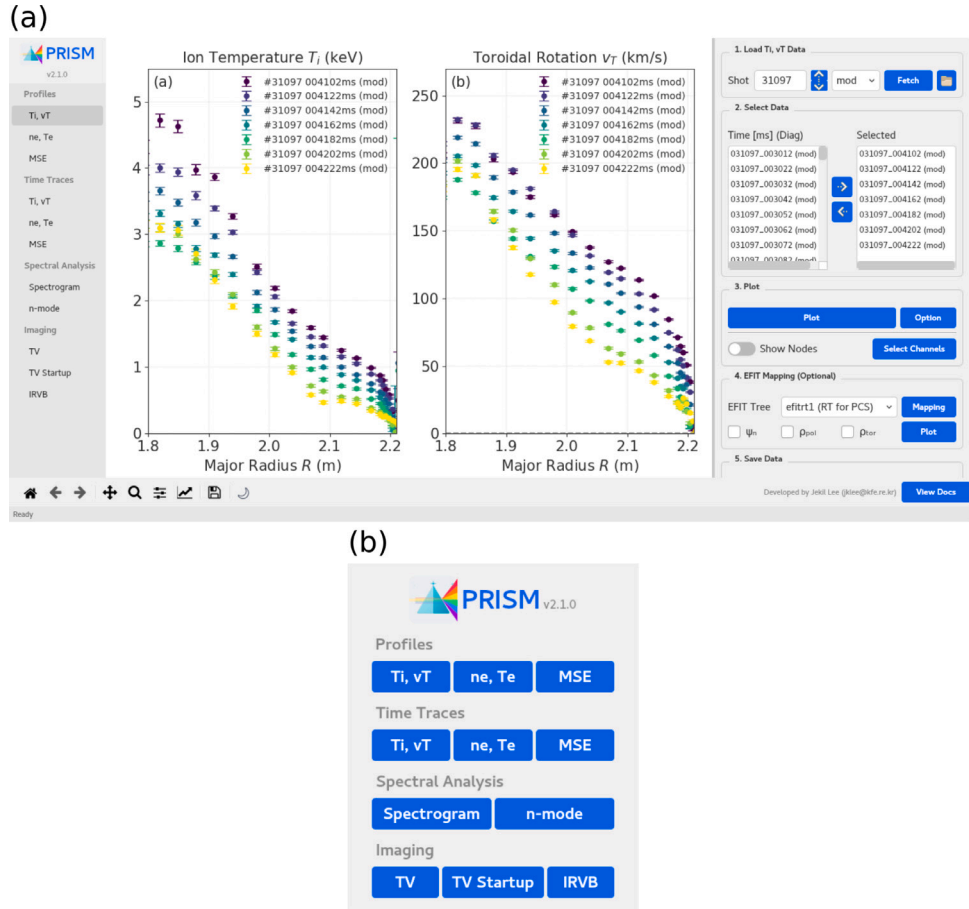


Fig. 2. (a) PRISM graphical user interface. A categorized sidebar on the left groups diagnostic viewers into Profiles, Time Traces, Spectral Analysis, and Imaging categories. The center panel displays the plot canvas and the right panel contains control sections. (b) Tab selector mode presenting a lightweight launch menu for opening individual viewers. Both panels show the light theme.

2.2. On-demand tab initialization and standalone execution

To minimize startup time, PRISM employs on-demand tab initialization. When the application launches, only tab configurations are registered for each diagnostic tab. The full tab content — including data loaders, matplotlib figures, and control widgets — is instantiated only when the user first selects that tab. This approach significantly reduces the initial startup time compared to creating all tabs simultaneously, while maintaining fast response time for subsequent tab switching.

Additionally, PRISM supports standalone execution of individual diagnostic viewers through the tab selector mode (`prism -s`), which presents a categorized list of all available viewers as a lightweight launch menu (Fig. 2(b)). Users can open individual viewers in separate windows without loading the full platform.

2.3. Data flow

Diagnostic data is retrieved from the MDSplus server through dedicated loader modules. Each diagnostic has a dedicated loader that retrieves data and returns it in a standardized format, providing a uniform interface for accessing measurements regardless of the underlying data structure.

For high-frequency diagnostics such as Mirnov coils, BES, TCI, ECE, and ECEI, raw signals can contain millions of samples, making direct retrieval impractical due to memory constraints and access time. PRISM addresses this using MDSplus server-side functions such as `SetTimeContext` and `resample` to retrieve only the relevant portion of each

signal at the desired sampling rate, minimizing memory usage and data access time. After data retrieval, the time context is reset to avoid affecting subsequent queries.

For diagnostics that support file-based input, a separate parser module handles file reading and converts the data into the same format, allowing the visualization and analysis code to be used regardless of the data source. EFIT data is loaded through a dedicated loader that retrieves flux surface information and provides interpolation functions for coordinate transformation.

3. Supported diagnostics

The current version of PRISM integrates eleven diagnostic systems for comprehensive plasma characterization on KSTAR. Spatially resolved profile diagnostics — CES (Section 3.1), XICS (Section 3.2), TS (Section 3.3), ECE (Section 3.4), and MSE (Section 3.6) — provide radial measurements of plasma parameters. TCI (Section 3.5) provides line-integrated electron density. For two-dimensional measurements, IRVB (Section 3.7) provides radiation profiles and TV (Section 3.8) provides visible camera images. Spectral analysis tools include spectrogram visualization for ECE, ECEI, BES, TCI, and Mirnov signals (Section 3.9), and toroidal mode number identification (Section 3.10).

3.1. Charge exchange spectroscopy (CES)

CES measures ion temperature (T_i) and toroidal rotation velocity (v_T) through spectroscopic analysis of fully ionized carbon (CvI) charge exchange emission. The system provides 32 spatially resolved

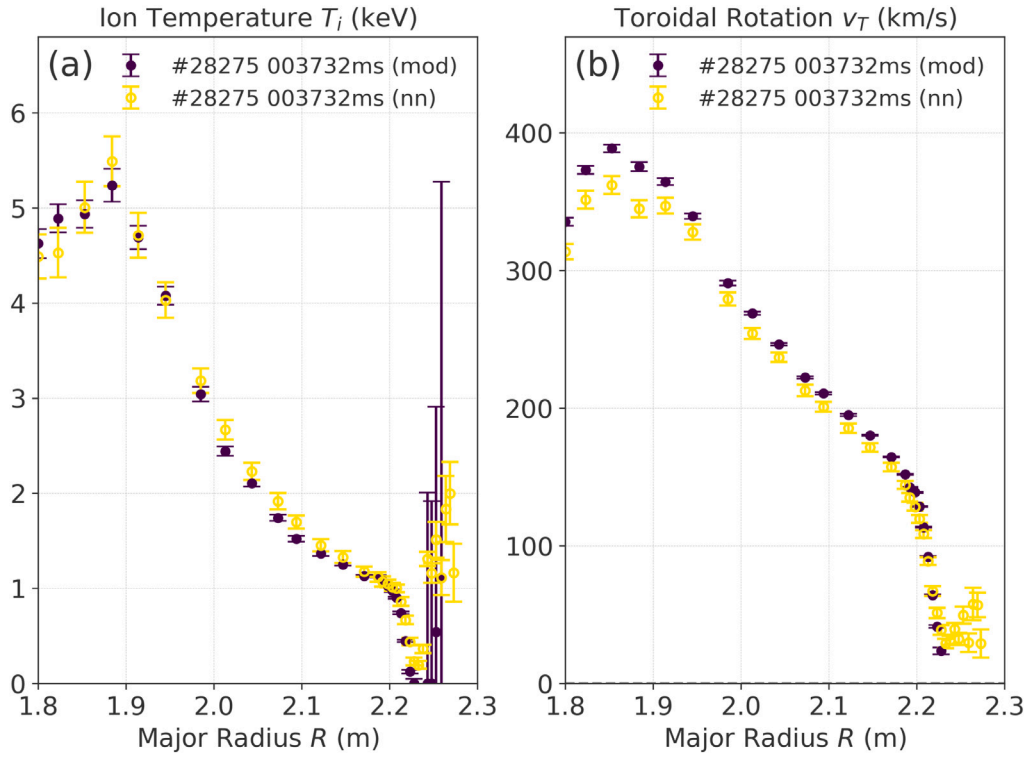


Fig. 3. (a) Ion temperature T_i and (b) toroidal rotation velocity v_T profiles are shown as a function of major radius R for #28275 at $t = 3732$ ms. Results from beam modulation (mod, purple) and neural network (nn, yellow) analysis methods are compared.

channels covering the plasma core to edge, with a time resolution of 10 ms.

PRISM supports two CES analysis methods selectable via a drop-down menu. The beam modulation method extracts active charge exchange signals by subtracting interpolated background emission estimated from periodic beam-off phases. Alternatively, the neural network method directly infers T_i and v_T from raw spectra without requiring beam modulation [21]. PRISM can also open CES analysis result files not uploaded to MDSplus, such as two-Gaussian fitting [22]. Fig. 3 shows profiles from both beam modulation and neural network methods for the same discharge.

3.2. X-ray imaging crystal spectrometer (XICS)

XICS measures T_i and v_T near the magnetic axis through X-ray spectroscopy of argon line emission. The system utilizes a double crystal assembly to simultaneously measure helium-like argon (Ar_{XVII}) and hydrogen-like argon (Ar_{XVIII}) spectra. Under high electron temperature conditions above ~ 5 keV, Ar_{XVIII} measurements are used since Ar_{XVII} atomic states are strongly depleted at such temperatures.

Validation studies have demonstrated that XICS and CES measurements show reasonable agreement in the core region despite measuring different impurity species [23]. Since the absolute wavelength is not calibrated for the KSTAR XICS system, the measured toroidal rotation is a relative velocity. PRISM automatically applies a v_T offset correction by comparing XICS and CES values at their first overlapping time point, facilitating direct comparison between the two diagnostics.

Fig. 4 shows the temporal evolution of core T_i and v_T from both XICS and CES during an ohmic discharge. The CES measurements were obtained using short beam blips to minimize external momentum input for intrinsic rotation studies in KSTAR [24].

3.3. Thomson scattering (TS)

The TS system provides measurements of electron temperature (T_e) and density (n_e) profiles through the scattering of laser light by free

electrons. PRISM loads data from core and edge channels, with channel positions configured in an external file according to shot number ranges to account for system reconfigurations over different experimental campaigns. The loader automatically handles the asymmetric error bars (upper and lower uncertainty bounds) provided by the TS analysis.

Fig. 5 shows electron temperature profiles from TS and ECE alongside electron density from TS, demonstrating the multi-diagnostic overlay capability.

3.4. Electron cyclotron emission (ECE)

ECE radiometry measures electron temperature through thermal emission at electron cyclotron harmonics. The ECE diagnostic on KSTAR provides high temporal resolution measurements suitable for studying fast phenomena. The raw ECE signals are acquired at high sampling rates (500 kHz by default, and up to 2 MHz), making direct retrieval impractical. PRISM utilizes the built-in `SetTimeContext` function of MDSplus to retrieve resampled data within the specified time range, with configurable display rates of 100 Hz to 1 kHz.

A key feature of the ECE implementation is the automatic handling of harmonic overlap. ECE signals originate from specific radial locations determined by the magnetic field strength and emission frequency. When both second and third harmonic emissions fall within the analysis region on the lower field side, the third harmonic can contaminate the second harmonic signal. PRISM automatically identifies channels where this overlap occurs and marks them with distinct markers in the visualization, allowing researchers to distinguish reliable measurements from potentially contaminated ones.

Additionally, baseline correction is automatically applied by subtracting the average signal during the pre-discharge phase from the entire time trace. This removes DC offsets and improves the accuracy of temperature measurements.

Fig. 6(a) shows the temporal evolution of T_e from ECE and TS, demonstrating the multi-diagnostic time trace comparison capability.

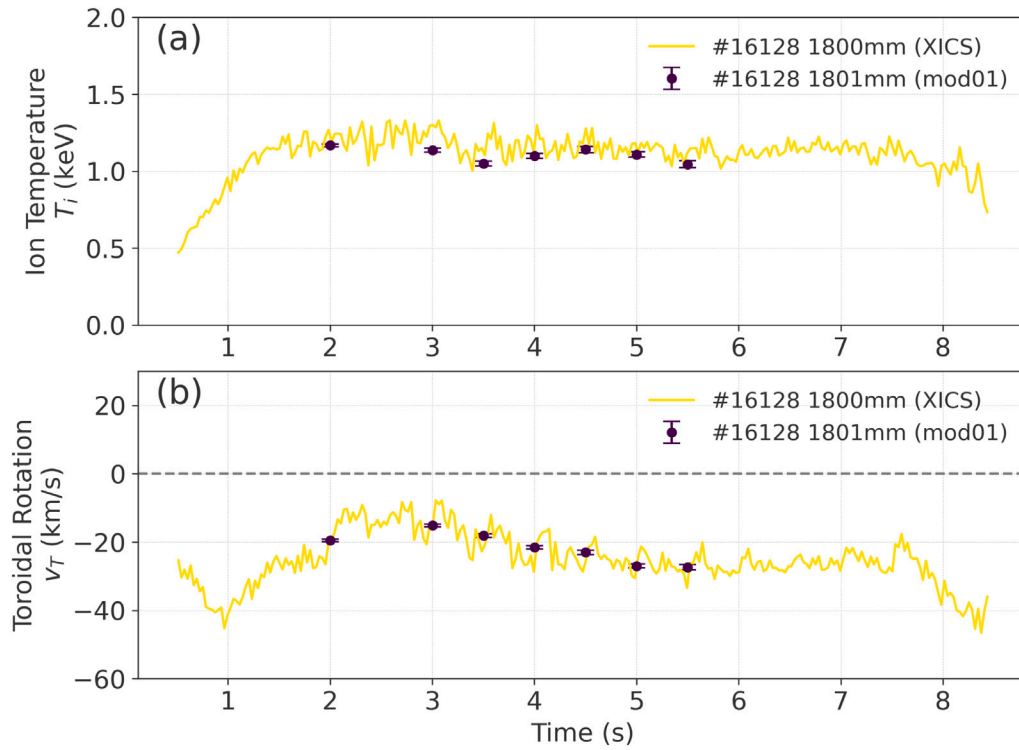


Fig. 4. Temporal evolution of (a) T_i and (b) v_T from XICS (yellow) and CES (purple, beam modulation analysis) near the magnetic axis ($R = 1.80$ m) for #16128. The sparse CES data points result from short 10 ms beam blips during this ohmic discharge. The v_T offset between XICS and CES is automatically corrected.

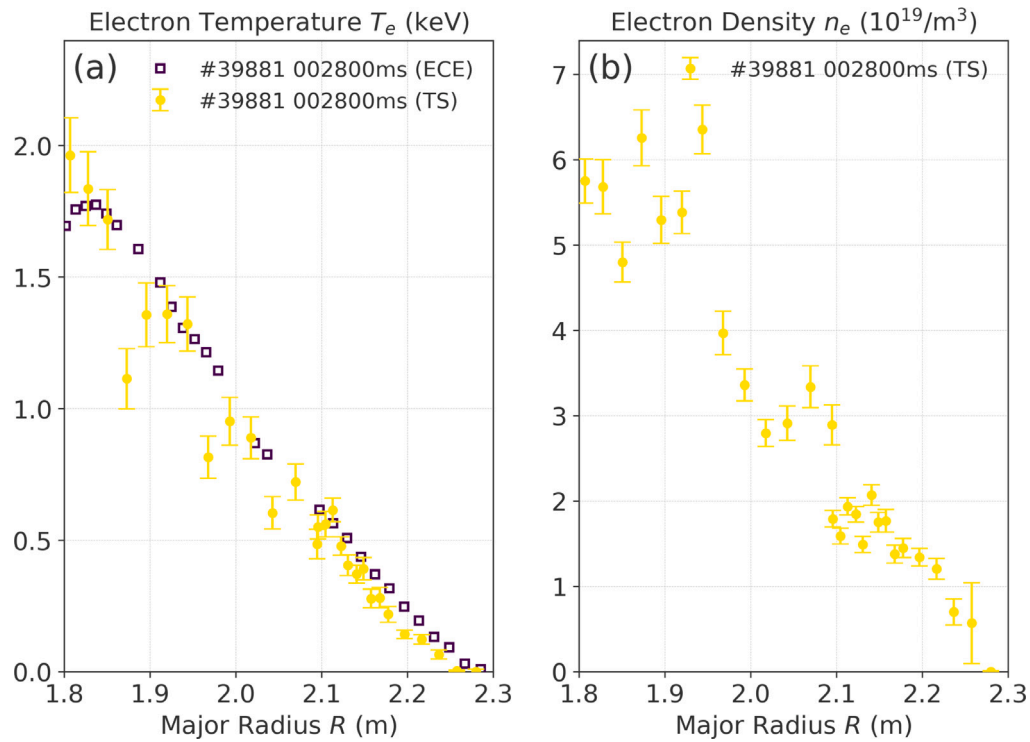


Fig. 5. (a) Electron temperature T_e and (b) density n_e profiles are shown as a function of major radius R for #39881 at $t = 2800$ ms. In (a), T_e from TS (yellow, filled circles with error bars) and ECE (purple open squares) are compared. In (b), n_e from TS is shown.

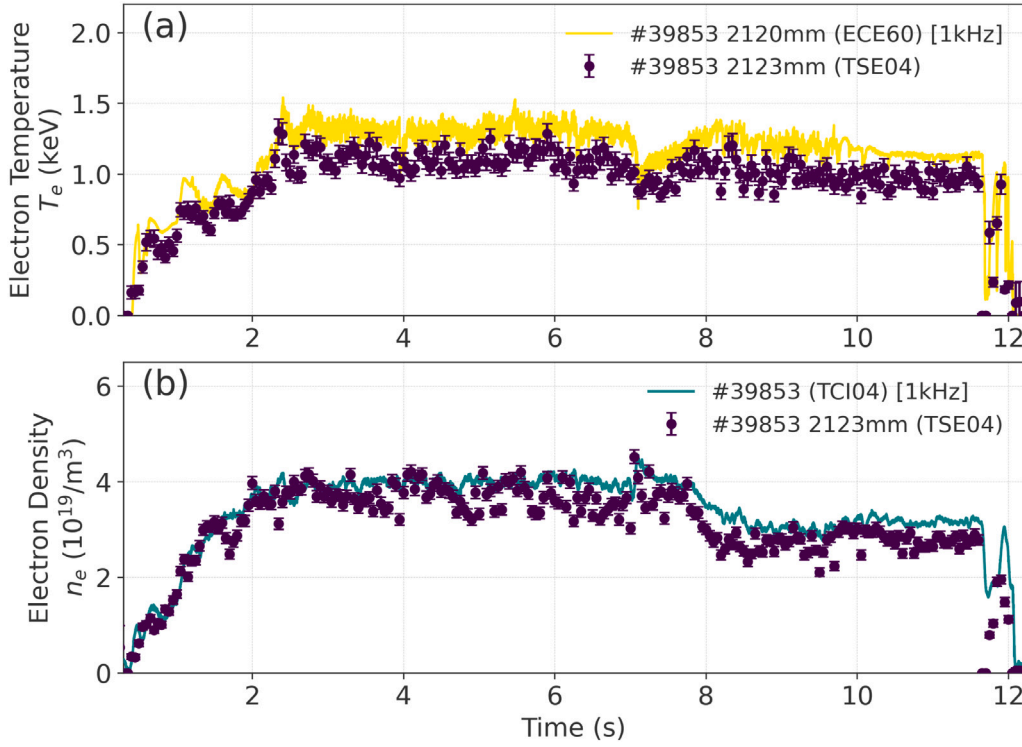


Fig. 6. Temporal evolution of T_e and n_e for #39853. (a) T_e at $R = 2.12$ m from ECE (yellow) and TS (purple). (b) Line-integrated n_e from TCI channel 4 (teal) with local n_e from TS at $R = 2.12$ m (purple). ECE and TCI signals are displayed at 1 kHz sampling rates. The resonant magnetic perturbation (RMP) coil current was gradually applied from $t = 6$ s, and density pump-out is observed in (b).

3.5. Two-color interferometer (TCI)

TCI measures line-integrated electron density along five tangential chords passing through the plasma. The system provides high temporal resolution measurements suitable for tracking density evolution during transient events. PRISM displays the temporal evolution of line-integrated density from all five channels, with selectable sampling rate of 100 Hz for overview or 1 kHz for detailed analysis while the original signal provides up to 625 kHz sampling rates.

Fig. 6(b) shows the line-integrated density from TCI channel 4 alongside local n_e from TS at $R = 2.12$ m during an RMP experiment, where density pump-out is observed as the RMP current increases from $t = 6$ s.

3.6. Motional Stark effect (MSE)

MSE diagnostics measure the magnetic pitch angle (γ) through polarization analysis of Doppler-shifted Balmer-alpha emission from neutral beam atoms. The pitch angle measurements are typically used as constraints for equilibrium reconstruction codes such as EFIT (see Section 4.3). In addition, the safety factor (q) and current density (j) profiles can be derived directly using simplified flux surface models without requiring full equilibrium reconstruction [25]. Based on this approach, PRISM displays γ measurements in one panel and the derived q or j profile in another panel.

The MSE loader automatically determines the beam source from the diagnostic configuration. Bad channels are automatically detected and excluded from the analysis.

Fig. 7 shows γ and j profiles with on-axis and off-axis NB current drive. The on-axis case exhibits a peaked j profile with higher core current density, while the off-axis cases show flatter profiles with reduced core values, reflecting the different current deposition locations.

3.7. Infrared video bolometer (IRVB)

IRVB measures the two-dimensional plasma radiation distribution using a pin-hole camera with a platinum foil screen and an infrared (IR) camera [17]. The foil temperature changes induced by plasma radiation are measured by the IR camera and converted into radiation power, and the two-dimensional plasma emissivity profiles are reconstructed through tomographic inversion.

PRISM loads the reconstructed radiation data from the IRVB analysis server and provides visualization with EFIT overlay from selectable EFIT trees. The viewer displays the 2D radiation distribution in the poloidal cross-section with flux surface contours and plasma boundary. Users can define normalized poloidal flux boundaries to separate the plasma into regions such as core, edge, and scrape-off layer, and PRISM computes the radiated power for each region by integrating the emissivity weighted by the toroidal volume element. The temporal evolution of these regional radiated powers is displayed as time traces alongside the 2D emissivity distribution. Fig. 8 compares upper single null (USN) and lower single null (LSN) configurations. With the full tungsten tiles installed in the lower divertor after the 2023 experimental campaign in KSTAR, the USN discharge shows radiation concentrated near the upper carbon divertor, while the LSN discharge exhibits strong core radiation characteristic of tungsten impurity accumulation.

3.8. Visible camera (TV)

TV captures plasma images for visual inspection of plasma behavior and wall interactions. PRISM provides a dedicated viewer that loads image sequences by shot number, automatically detecting available camera data.

Similar to the IRVB viewer, frame-by-frame navigation and playback controls enable examination of plasma dynamics. A draw mode allows users to mark line segments on the image, marking plasma boundaries or other features of interest directly on the displayed images. The drawn

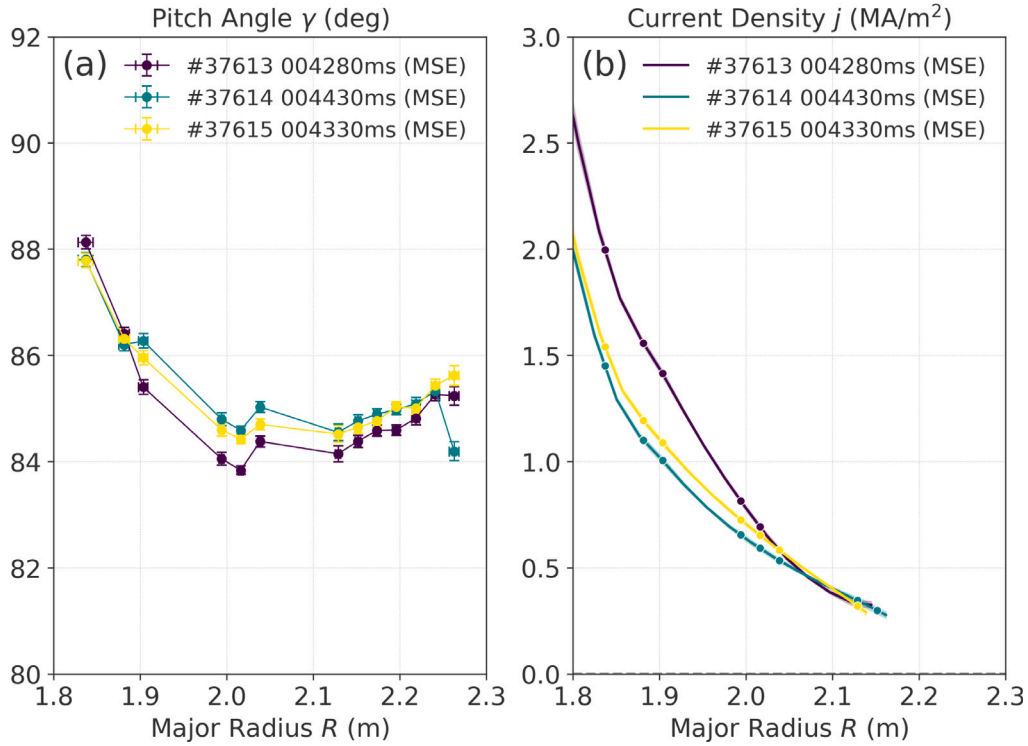


Fig. 7. (a) Magnetic pitch angle γ and (b) current density j profiles are shown as a function of major radius R for three discharges with different NB current drive configurations: #37613 with on-axis NB2 A source (purple), #37614 with off-axis NB2B source (teal), and #37615 with off-axis NB2C source (yellow).

line can be customized with different colors, widths, and optional spline smoothing, and is preserved across frame changes for comparison of plasma surface position between frames.

Fig. 9(a) shows a plasma image from #31097 with the region of interest marked. Fig. 9(b)–(c) demonstrate consecutive frames capturing plasma surface displacement in the upper outboard region, which can be utilized for quantitative analysis of plasma boundary displacement such as edge kink-like modes observed during RMP application [26].

PRISM also includes a TV startup viewer for comparing plasma startup sequences across multiple discharges. Users add shots to a list, select a visible camera and frame range with bidirectional time synchronization, and the viewer generates a montage where each row corresponds to a different shot and each column to a frame number. Fig. 10 shows TV images from the startup scenario optimization after the installation of full tungsten tiles in the lower divertor during the 2023 KSTAR campaign [27]: before optimization (#33074), during optimization (#33100), and the final scenario (#33201).

3.9. Spectrogram

For studying high-frequency fluctuations, PRISM provides a dedicated spectrogram tab supporting ECE, ECEI, BES, TCI, and Mirnov coil signals. The spectrogram is computed using the short-time Fourier transform with configurable number of fast Fourier transform (FFT) points (256 to 4096) and adjustable frequency and time ranges.

For ECE, individual radiometer channels can be selected with their corresponding positions determined by the toroidal magnetic field of the shot. ECEI provides imaging data acquired at three locations, with selectable vertical and radial channel indices, which enables localized analysis of electron fluctuations at specific poloidal and toroidal positions. BES channels can be selected by vertical and radial indices. TCI offers both processed line-integrated density signals and raw interferometer signals for detailed fluctuation studies. For Mirnov coils, both toroidal and poloidal array configurations are supported. The dynamic range of the spectrogram display can be adjusted from 1 to 11 decades

using a slider control, allowing users to reveal weak fluctuations or focus on dominant modes.

Fig. 11 shows spectrograms from Mirnov coil and ECE measurements for a long-pulse discharge exhibiting long-lived pressure-driven MHD activity [28]. The dominant $n = 1$ mode and its harmonic are clearly visible throughout the plasma duration.

3.10. Toroidal mode number spectrum

Beyond basic spectrogram visualization, PRISM includes n identification capability for Mirnov coil data. The analysis uses the toroidal array of Mirnov coils to determine the dominant toroidal mode numbers present in magnetic fluctuations. Mirnov coil signals are sampled at 2 MHz, producing large datasets that require efficient handling. Data retrieval uses `SetTimeContext` to limit the time range on the server side and loads all channels in parallel to reduce access time. Since the toroidal coil array configuration — including channel polarity and sign conventions — has changed across experimental campaigns, these parameters are maintained in an external configuration file indexed by campaign year.

The algorithm performs an FFT on each Mirnov coil channel, then fits the phase variation across the toroidal array to identify n . Parameters including tolerance threshold, amplitude fraction, and number of modes to search are adjustable through the interface. The results are displayed as a color-coded spectrogram where each color represents a different mode number, along with the temporal evolution of mode amplitudes.

Fig. 12 presents the n spectrum analysis for #31747, where the neoclassical tearing mode (NTM) was stabilized using a minimum seeking feedback controller with electron cyclotron (EC) injection [29]. The $n = 1$ mode amplitude decreases significantly after $t = 7.3$ s when the EC launching angle was properly aligned with the NTM location.

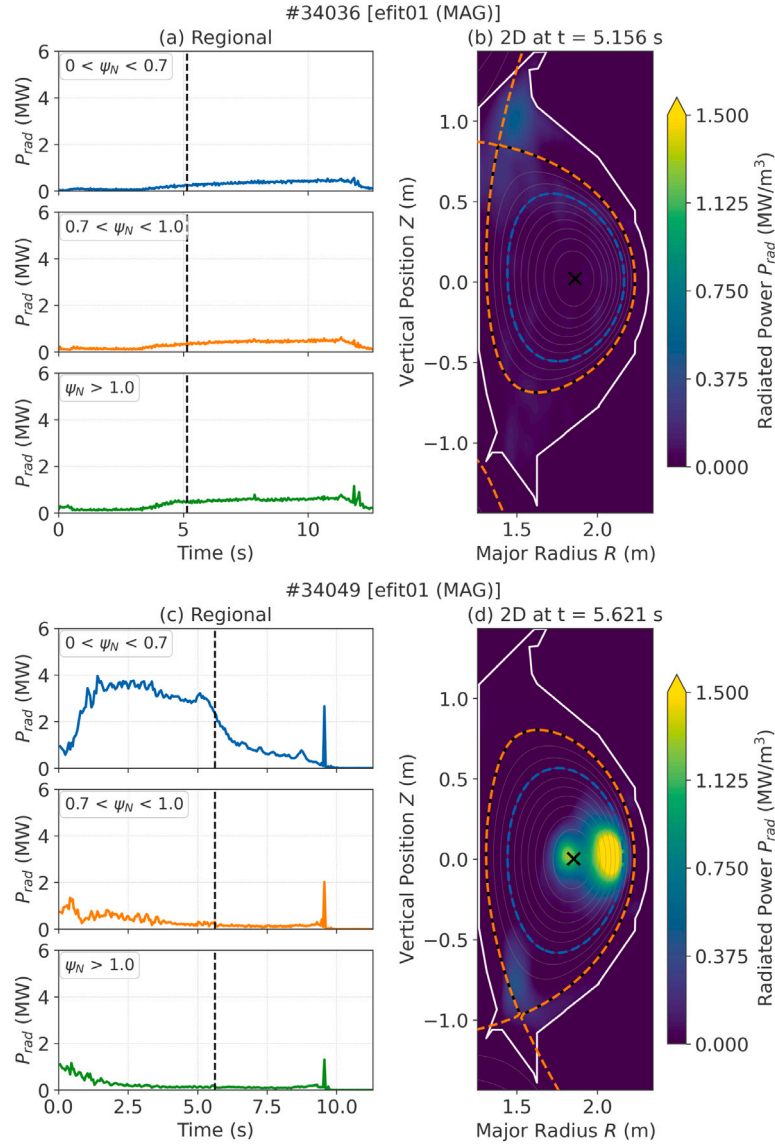


Fig. 8. IRVB viewer comparing upper single null (USN, #34036) and lower single null (LSN, #34049) configurations. (a), (c) Regional temporal evolution of radiated power P_{rad} for three plasma regions defined by normalized poloidal flux (ψ_N) boundaries: core ($0 < \psi_N < 0.7$, blue), edge ($0.7 < \psi_N < 1.0$, orange), and SOL ($\psi_N > 1.0$, green). Vertical dashed lines indicate the time slices shown in (b) and (d). (b), (d) Two-dimensional emissivity profiles at $t = 5.156$ s and $t = 5.621$ s, respectively. Blue and orange dashed lines mark the $\psi_N = 0.7$ and 1.0 boundaries, respectively, the white solid line indicates the limiter, and the black cross symbol denotes the magnetic axis. The USN case shows radiation near the upper carbon divertor, while the LSN case exhibits enhanced core radiation from tungsten impurities originating from the lower tungsten divertor.

4. Key features

4.1. Data quality control

It is important to note that PRISM primarily visualizes data that has already been calibrated and processed by diagnostic-specific analysis codes. The calibrated results are stored in the KSTAR MDSplus database by the respective diagnostic teams, and PRISM retrieves these processed results rather than performing raw data calibration itself. The diagnostic-specific knowledge incorporated into each data loader — such as channel configurations, data quality criteria, and coordinate conventions — was developed through discussions with the respective diagnostic operators at KSTAR.

PRISM implements several data quality control measures to ensure reliable visualization. All diagnostic data loaders automatically detect the plasma current (I_p) and mask data outside the valid plasma

phase, including only data when I_p exceeds 100 kA to filter out pre-discharge noise and post-disruption artifacts. If a shot fails to achieve $I_p > 100$ kA, the loader raises an informative error message. Bad channels are automatically identified and excluded: for MSE, channels with characteristic bad-channel signatures are detected, while for ECE, channels with zero mean signal are filtered out. Additionally, baseline and offset corrections are applied automatically—ECE signals are baseline-corrected using pre-discharge data, and XICS v_T offset is corrected by comparison with CES measurements. These automatic quality controls help prevent common pitfalls such as displaying data from failed shots or including contaminated channels, reducing the risk of misinterpretation compared to accessing raw MDSplus data directly.

4.2. Multi-shot, multi-timepoint, and multi-channel comparison

One of the key strengths of PRISM is the ability to simultaneously compare data from multiple shots, time points, and spatial channels.

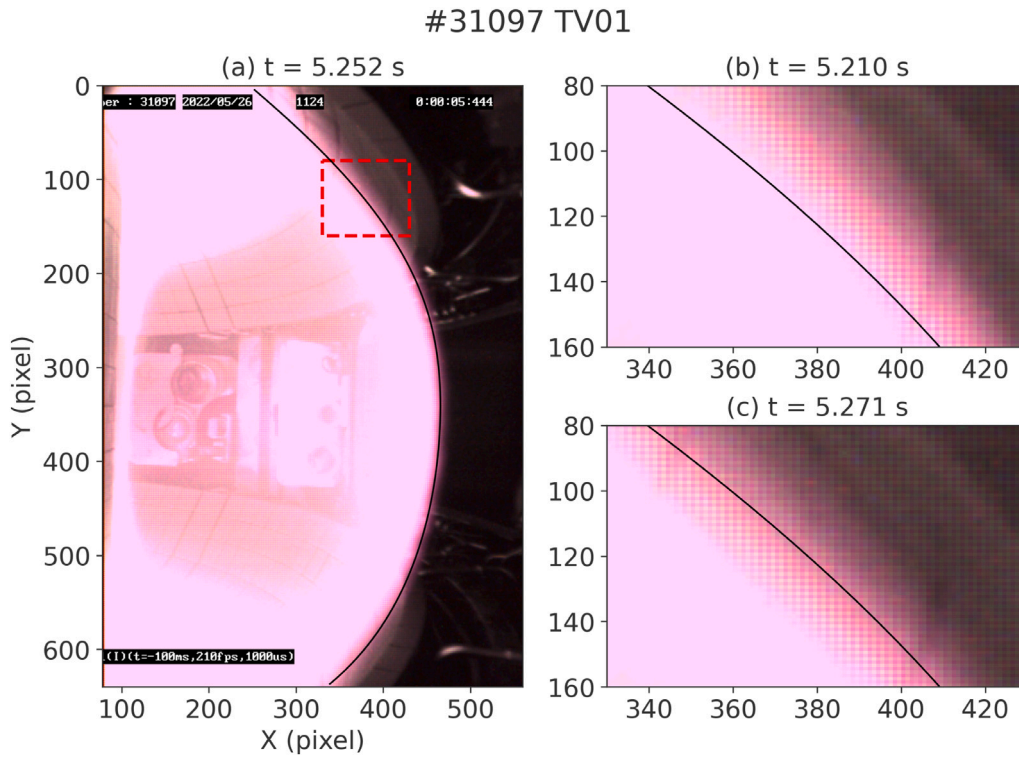


Fig. 9. TV viewer displaying #31097. (a) Plasma image with frame information overlay. The black solid line, drawn using the built-in line drawing feature, marks the plasma surface position. (b) and (c) show two frames of the upper outboard region indicated by the red dashed box in (a), capturing plasma surface displacement caused by edge kink-like modes [26].

Researchers frequently need to compare plasma profiles across different discharge conditions—before and after a transition, between discharges with different heating powers, or across an entire experimental day to identify trends.

PRISM allows users to select multiple time slices from different shots and overlay them on a single plot with automatic color coding and legend generation. The selection interface supports flexible data organization: users can load data from multiple shots into a common list and select arbitrary combinations of shot-time pairs for instant visualization. For time trace views, multiple spatial channels can be displayed simultaneously, enabling direct comparison of temporal evolution at different radial positions within a single shot or across multiple shots. This multi-shot, multi-timepoint, and multi-channel capability distinguishes PRISM from single-shot visualization tools and streamlines research workflows that would otherwise require manual data extraction and external plotting software. The workflow is particularly efficient during experimental campaigns where rapid assessment of parameter dependencies is essential for planning subsequent discharges. Multi-timepoint comparison is demonstrated in Figs. 3 and 5, multi-channel comparison in Figs. 4 and 6, and multi-shot overlay in Figs. 7 and 10.

4.3. Equilibrium mapping

One of the primary motivations for developing PRISM was to simplify the process of mapping diagnostic data to magnetic flux coordinates. The platform supports five EFIT trees available on KSTAR, including two real-time reconstructions used for the plasma control system, as summarized in Table 1.

For each EFIT tree, three flux coordinate systems are available: normalized poloidal flux, poloidal ρ (the square root of normalized poloidal flux), and toroidal ρ (the square root of normalized toroidal flux). Users can select the EFIT tree and coordinate system through the

Table 1
EFIT trees supported by PRISM.

Tree	Type	Description
efitrt1	Real-time	Magnetic constraints
efitrt2	Real-time	Magnetic + MSE constraints
efit01	Post-discharge	Magnetic constraints
efit02	Post-discharge	Magnetic + MSE constraints
efit04	Post-discharge	Magnetic constraints with drift correction

interface, and the platform automatically performs the interpolation from major radius to the selected flux coordinate.

Fig. 3 shows ion temperature and toroidal rotation velocity profiles as a function of major radius from both beam modulation and neural network analysis methods. Fig. 13 demonstrates the same data mapped to toroidal ρ using the `efitrt1` equilibrium reconstruction. This mapping is essential for comparing profiles across different plasma conditions where the magnetic geometry varies.

4.4. Data export

PRISM provides data export capabilities for further analysis. The navigation toolbar allows saving figures as image files. For profile and time trace tabs, selected data can be exported to comma-separated value (.csv) files with EFIT coordinate information included. Before saving, a tabular data preview dialog displays the formatted data in a spreadsheet-style view, allowing users to verify the contents before export.

For analysis-intensive tabs (spectrogram, n -mode spectrum, and IRVB), PRISM exports data in .npz format, a compressed archive format from NumPy, a Python numerical computing library. The .npz files contain processed results along with axis arrays (time, frequency, radial coordinates) and analysis parameters, enabling users to reproduce figures or perform additional analysis in external tools.

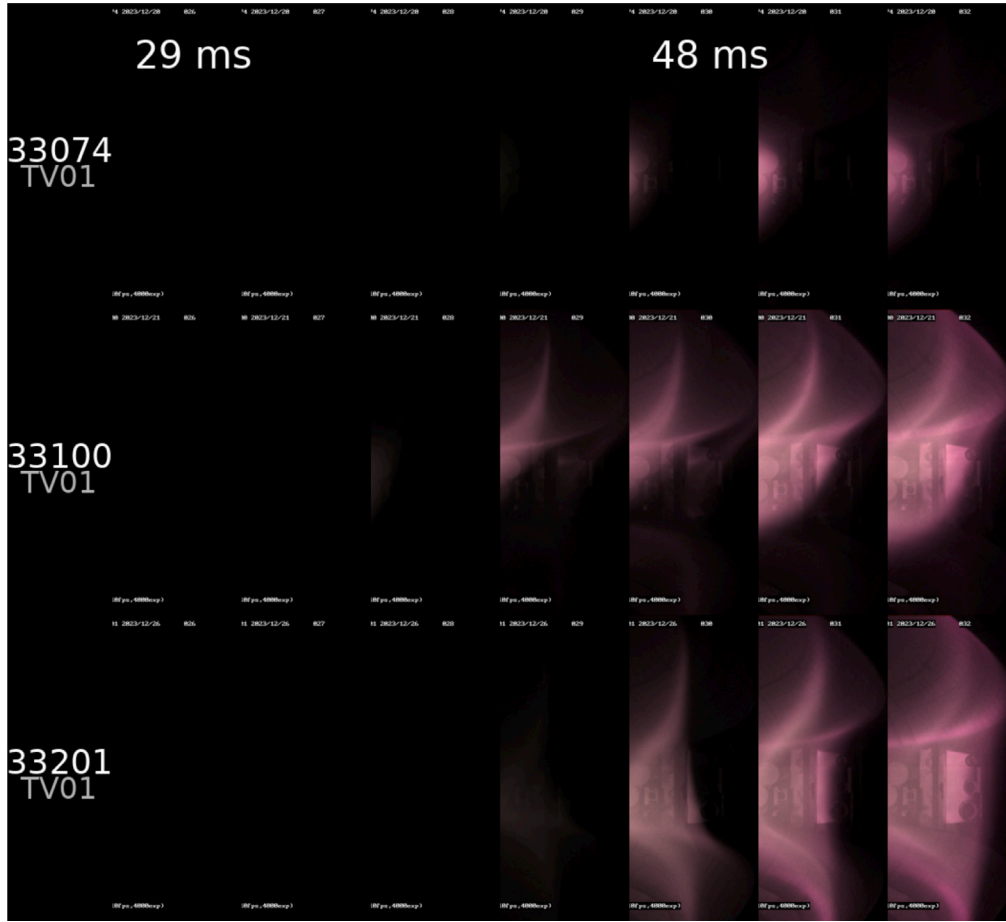


Fig. 10. TV startup viewer displaying images from #33074 (before optimization), #33100 (during optimization), and #33201 (final scenario) during the 2023 KSTAR campaign with the newly installed lower tungsten divertor [27]. Each row corresponds to a different discharge and each column to a frame number.

4.5. Plot customization

All visualization tabs allow users to customize plot appearance, including color schemes using matplotlib colormaps and font sizes for axis labels, legends, and tick marks. These settings are automatically persisted across sessions, so user preferences are restored on subsequent launches.

5. Discussion

Various visualization tools have been developed in the fusion community for diagnostic data access. MDSplus-based tools such as jScope [19] and π Scope [30] provide general-purpose waveform display and multi-discharge plotting capabilities. JETDSP [31] offers interactive signal analysis at JET. ReviewPlus and EFITviewer [32], developed at DIII-D, provide diagnostic signal visualization and EFIT equilibrium reconstruction display, respectively. EASTprofiles [33] provides profile visualization with equilibrium mapping for EAST tokamak, and MyView2 [34] at LHD supports cross-shot analysis with offline data caching for remote collaboration. Within the ITER Integrated Modeling and Analysis Suite (IMAS) [35], IMASviz [36] provides visualization of data stored in IMAS Interface Data Structures (IDSs), and IMAS-ParaView [37] extends ParaView [38], an open-source scientific visualization platform, to display IMAS data including generalized grid description structures. The OMFIT framework [39] includes the SCOPE module [40] for comparing time-series signals across discharges on a common timebase.

PRISM was developed to address specific workflow requirements at KSTAR, where researchers previously needed to manually extract data

and perform coordinate transformations using external scripts for cross-diagnostic comparison. Beyond profile visualization with equilibrium mapping, PRISM integrates a broader range of diagnostics including MHD fluctuation analysis and two-dimensional imaging under a unified interface. The multi-shot, multi-timepoint, and multi-channel overlay capability, combined with automatic color coding and legend generation, streamlines the comparison workflow that is essential during experimental campaigns.

Compared with the pre-existing tools at KSTAR, PRISM extends the single-diagnostic approach of CESviewer to eleven diagnostic systems under a unified interface, enabling overlay of diagnostics measuring the same parameter (CES and XICS for T_i and v_T , TS and ECE for T_e , TS and TCI for n_e). PRISM also broadens the Mirnov coil spectrogram tool from two signal types (Mirnov coil and ECE) to five (Mirnov coil, ECE, TCI, ECEI, and BES), and adds TV visible camera and IRVB radiation profile visualization. While jScope provides general-purpose MDSplus waveform display, PRISM adds diagnostic-specific processing including automatic EFIT coordinate mapping, bad channel filtering, and plasma current masking that are not available in general-purpose viewers.

The platform capabilities can support various physics studies on KSTAR. For example, the XICS and CES comparison functionality is relevant to intrinsic rotation studies [24], the TV viewer with line drawing functionality to edge stability studies [26], the TV startup viewer to plasma startup scenario optimization [27], the spectrogram to long-lived pressure driven MHD studies [28], and the n -mode spectrum analysis to NTM stabilization experiments [29].

In a typical analysis workflow, PRISM serves as the first step for data examination and selection. For detailed processing or profile

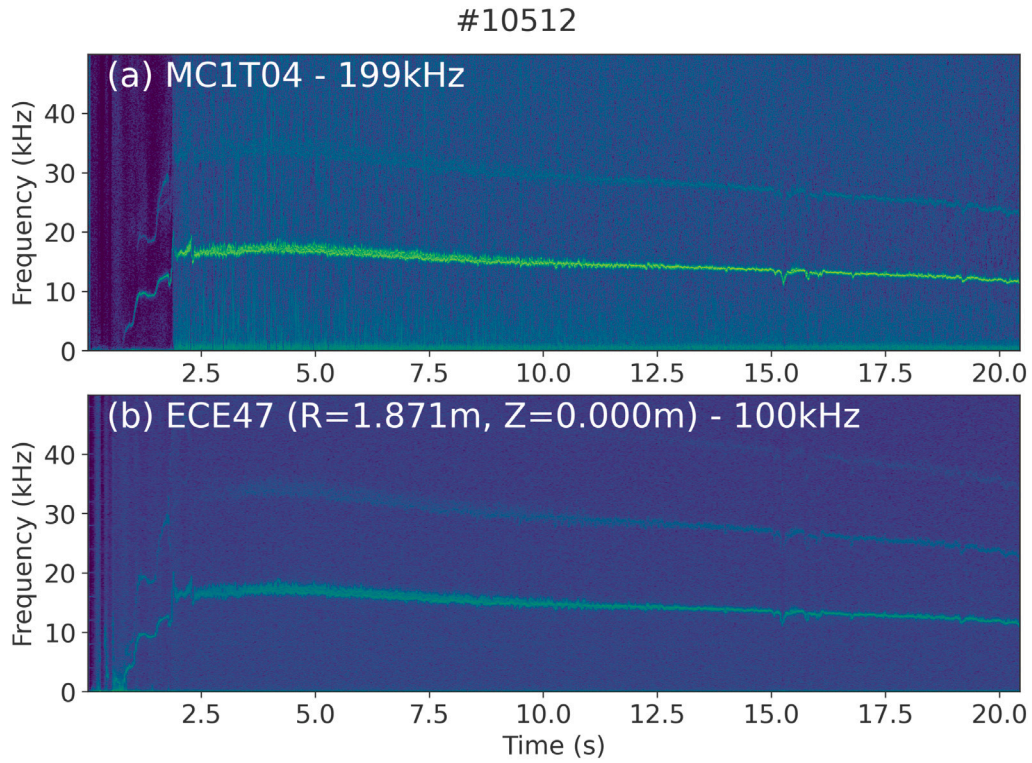


Fig. 11. Spectrograms of #10512 showing long-lived MHD activity [28]. (a) Mirnov coil toroidal array channel 4 signal. (b) ECE signal at $R = 1.87$ m. The dominant $n = 1$ mode and its harmonics persist throughout the discharge.

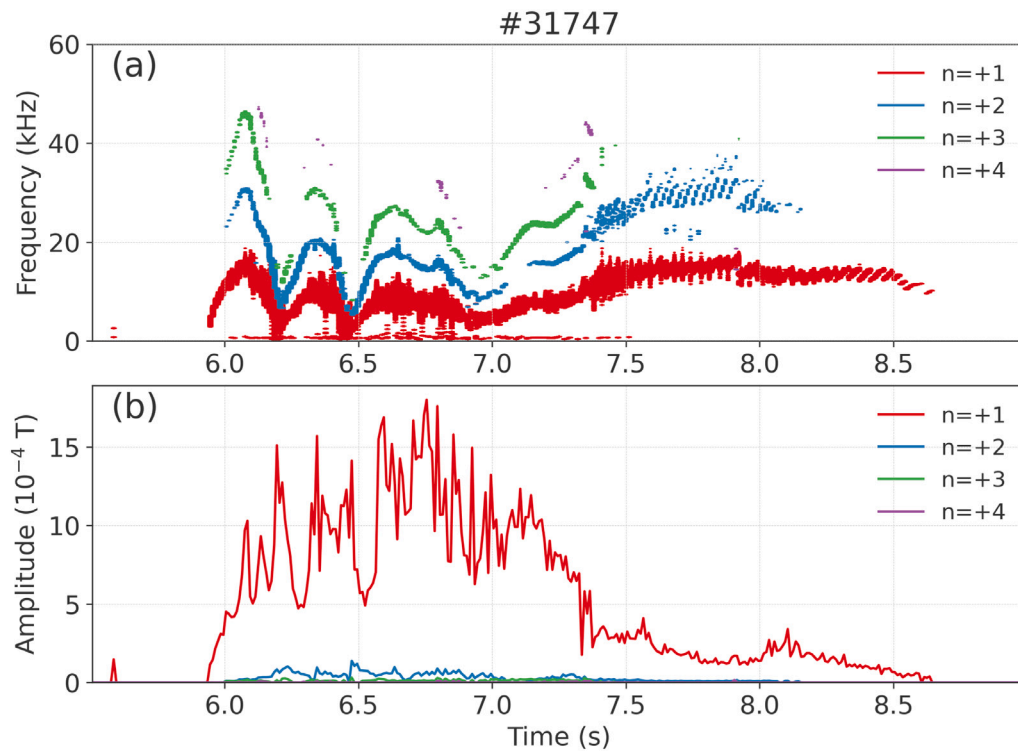


Fig. 12. Toroidal mode number spectrum analysis for #31747 during NTM stabilization experiments using minimum seeking feedback control [29]. (a) Mode frequency and (b) amplitude are shown as a function of time for each toroidal mode number n .

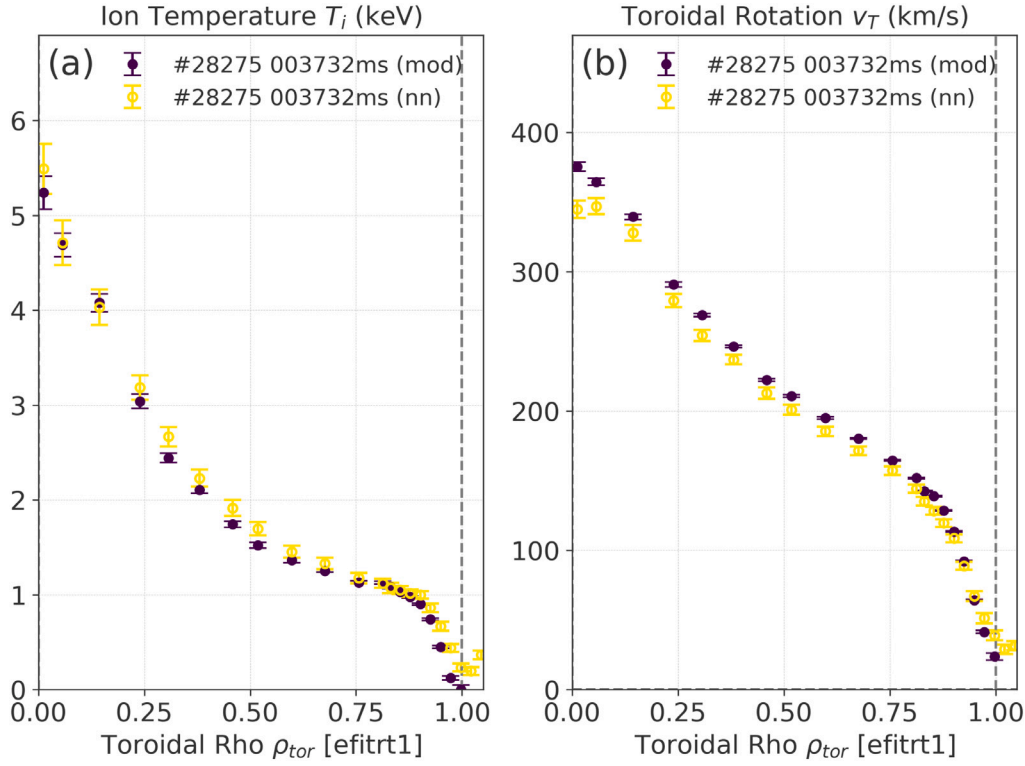


Fig. 13. (a) Ion temperature T_i and (b) toroidal rotation velocity v_T profiles mapped to toroidal rho ρ_{tor} using efitrt1 equilibrium for #28275 at $t = 3732$ ms. Results from beam modulation (mod, purple) and neural network (nn, yellow) analysis methods are shown. The gray dashed line indicates the separatrix ($\rho_{tor} = 1$).

fitting, dedicated tools on KSTAR such as OMFITprofiles [41] and GEFIT [42] are then utilized.

Among the supported viewers, the n -mode spectrum analysis is the most computationally demanding. A typical analysis uses approximately 15 toroidal Mirnov coil channels, each sampled at 2 MHz. For a 10 s analysis window, this amounts to 3×10^8 samples (~ 1.2 GB), which are loaded in parallel from the MDSplus server. The subsequent per-channel FFT computation and toroidal phase fitting at each frequency and time step further increase the memory footprint. The spectrogram tab faces similar data volume constraints when processing high-frequency diagnostics such as Mirnov coils, ECE, ECEI, and BES. Profile and time trace visualization, which constitute the majority of daily usage, retrieve pre-processed data at much lower sampling rates and are not subject to these constraints. PRISM is currently deployed on the KSTAR experimental server cluster, which provides sufficient computational resources for the analysis tasks described in this work.

As KSTAR continues to develop new diagnostic systems, the platform can accommodate them through its loader-based architecture described in Section 2: adding a new diagnostic requires implementing a loader module and registering it in the configuration without modifying the existing codebase. The same approach enables adaptation to other fusion devices, as the visualization and analysis components are independent of the specific data source.

6. Summary and future work

PRISM provides an integrated platform for multi-diagnostic data visualization on the KSTAR tokamak, currently unifying eleven diagnostic systems under a common interface with automatic EFIT mapping and data quality control. The platform features a categorized sidebar navigation with dark and light theme support, per-channel visibility controls, customizable plot appearance with persistent settings, and a tab selector mode for launching individual viewers. The modular architecture facilitates extension to additional diagnostics without modifying

the core framework. The platform has been operational on the KSTAR experimental server since 2025 and is actively used by researchers.

Key capabilities include: (1) unified multi-diagnostic visualization with automatic equilibrium mapping from selectable EFIT trees, (2) multi-shot, multi-timepoint, and multi-channel overlay with automatic color coding, (3) consolidated data quality controls across all diagnostics, (4) two-dimensional IRVB radiation profile visualization with equilibrium overlay and region-integrated radiated power calculation, and (5) TV visible camera viewer with line drawing annotation and frame-by-frame playback.

Future development plans include two-dimensional ECEI visualization for electron temperature fluctuation studies, as well as EFIT-based flux surface mapping on TV images. Additional diagnostics including neutron diagnostics, infrared TV (IRTV), and Langmuir probes are planned to be incorporated into the platform. Building on previous work that demonstrated plasma position estimation from TV images using toroidal symmetry [43], this feature will enable direct comparison of equilibrium reconstruction with visible camera observations. As KSTAR plans to provide Bayesian inference-based profile fitting results and automated between-shot TRANSP analysis results [44] through the MDSplus database, PRISM will be extended to visualize these results alongside the measured data. Furthermore, while the current implementation is tailored to KSTAR diagnostics, the modular architecture allows straightforward adaptation to other fusion devices by implementing device-specific data loaders and modifying configuration files.

Code availability

The PRISM source code is publicly available at <https://github.com/jekillee/PRISM>.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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